AAE 364 Control Systems Analysis
3 credits

1 Course Information/Description

Modeling and analysis of dynamical systems with applications. Laplace transforms, transfer functions, block diagrams. Transient and steady-state response of dynamical systems. Root Locus, Bode, Nyquist methods for control systems analysis. Introduction to controller design.

2 Learning Resources and Reference Text

Brightspace Learning Management System:

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook: K. Ogata, Modern Control Engineering, 5th Ed., Prentice Hall

3 Assignments

The course has weekly homework assignments.

4 Course Learning Objectives

The objectives of this course are to (1) learn tools for the analysis of dynamical systems, and (2) learn how to design controllers for dynamical systems, including aircraft and spacecraft examples.

- Introduction of concept of control and its application, including flight control for aircraft and spacecraft
- Analysis: Evaluation of the behavior of a dynamical system by analyzing and understanding its transient and steady-state response using Root Locus, Bode, and Nyquist methods
- Control design: Design a PID-type controller for a dynamical system so that the closed-loop system can satisfy the given performance specification

Main Topics

- Basic mathematical tools
  - Complex variables and functions
  - Laplace transformation,
  - Linear, time-invariant, differential equations
- Mathematical modeling of dynamical systems
- Control systems analysis and design
  - Time-domain analysis: Stability and transient and steady-state performance
  - Frequency domain analysis and design

5 Evaluation

All assignments must be submitted on BS or Gradescope as instructed. No email submissions will be accepted. No late homework will be accepted without the instructor's prior consent. Make sure to communicate with the instructor clearly and ASAP in case you anticipate a late submission.

Your grade will be calculated as follows:

- Attendance: 5%
- Homework Assignments: 25%
- Midterm Exams (two): 40% (20% each)
- Final Exam: 35%

Your course grade will be based on the following grading scale:

Letter Grade	Percentage
A+	97-105%
A	92-96%
A-	90-91%
B+	87-89%
B	84-86%
B-	80-83%
C+	77-79%
C	74-76%
C-	70-73%

Letter Grade	Percentage
D+	67-69%
D	64-66%
D-	60-63%
F	0-59%